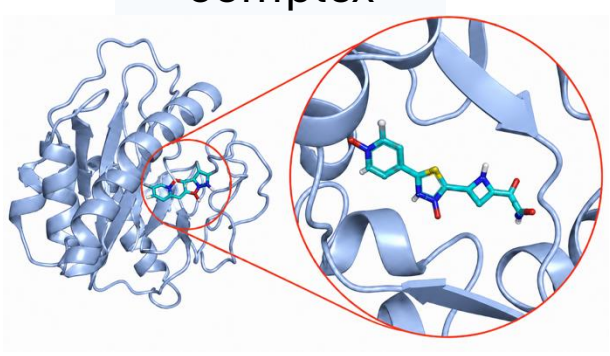
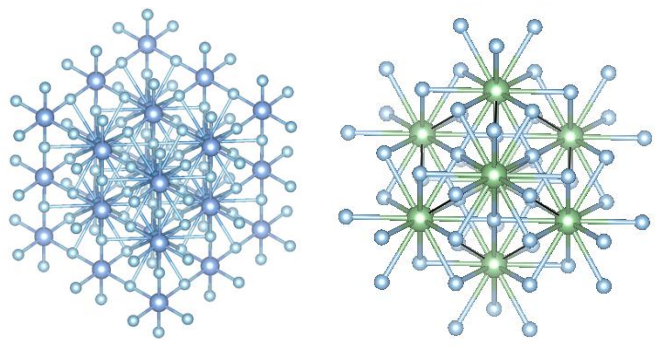
A Molecules & Materials

A1 Protein–ligand complex

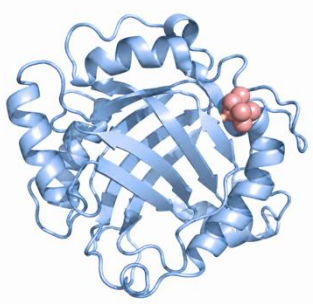


A2 MOF structure

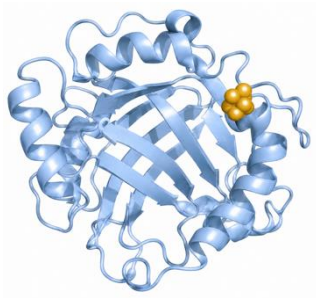


A3 Protein structure

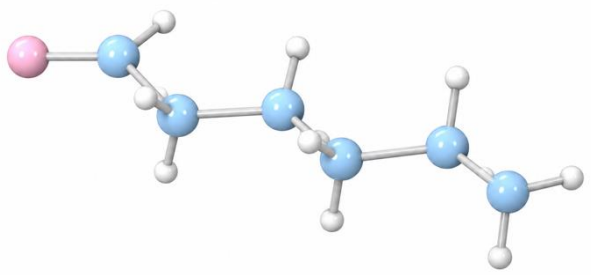
Wild



Mutant

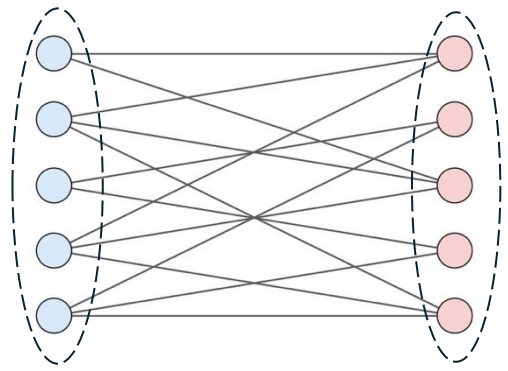


A4 Small molecules

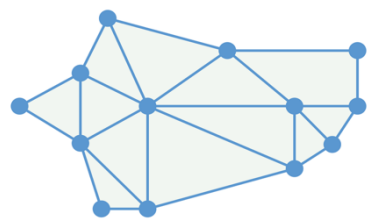


B Mathematical domains

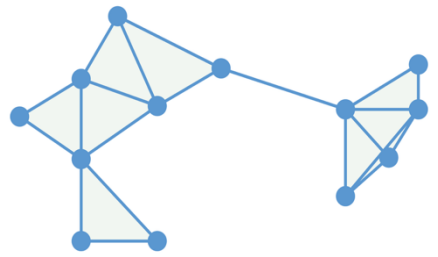
B1 Bipartite graph



B2 Simplicial complex

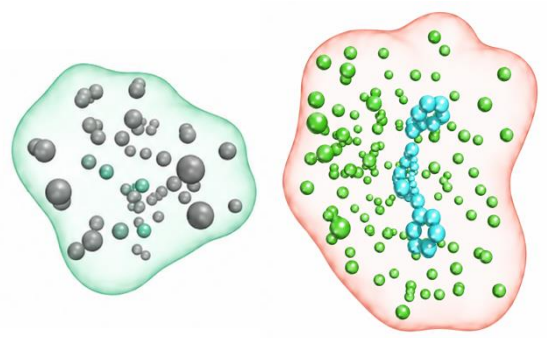


Vietoris-Rips complex



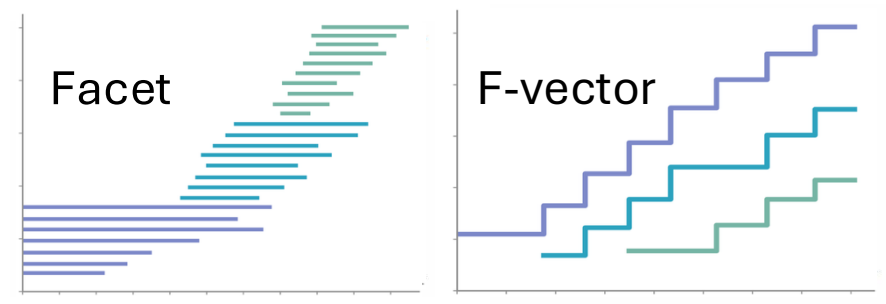
Alpha complex

B3 Differentiable manifold

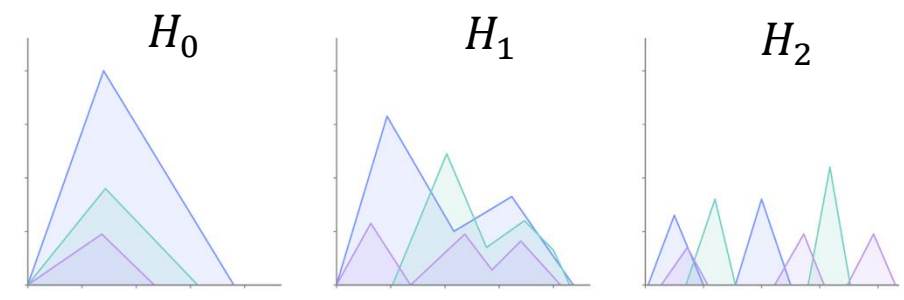


C Mathematical invariants

C1 Commutative algebra

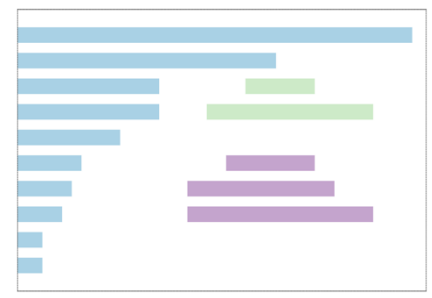


C2 Persistent homology

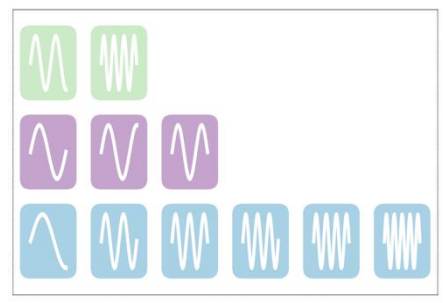


C3 Persistent Laplacian

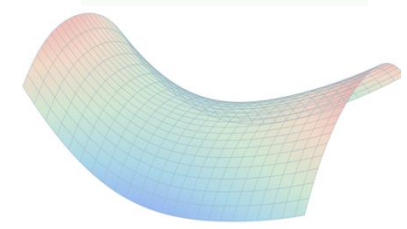
Harmonic spectra



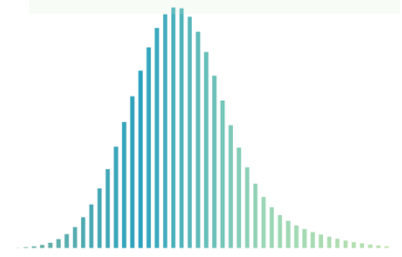
Nonharmonic spectra



C4 Element interactive curvature



C5 Forman persistent Ricci curvature

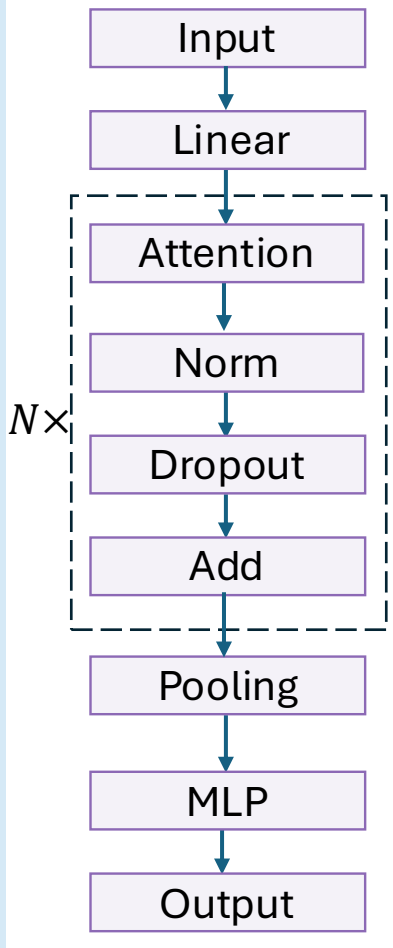


D Deep learning architectures

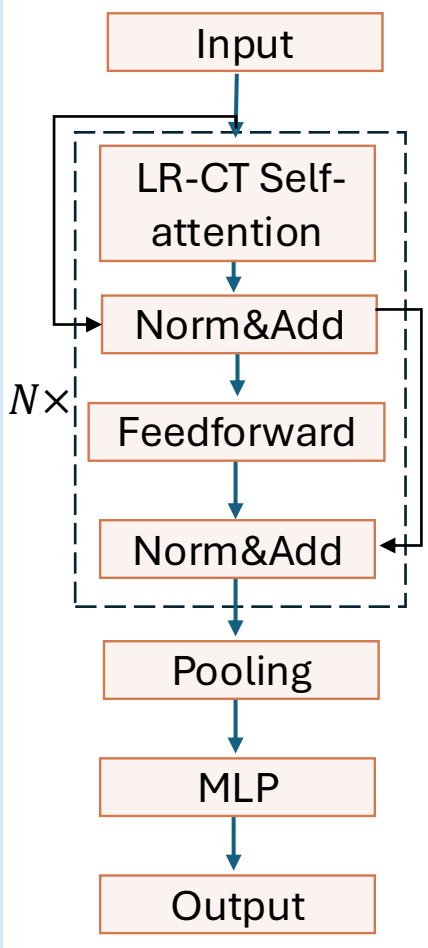
Topological neural architectures

Classic neural architectures

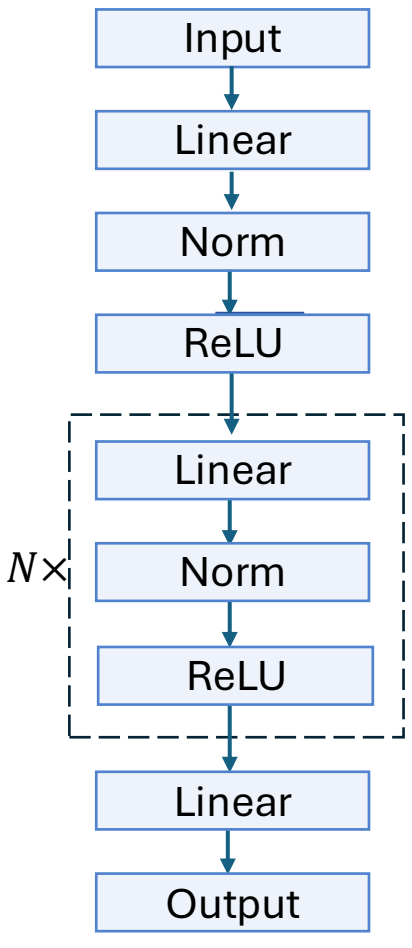
D1 Simplicial neural network



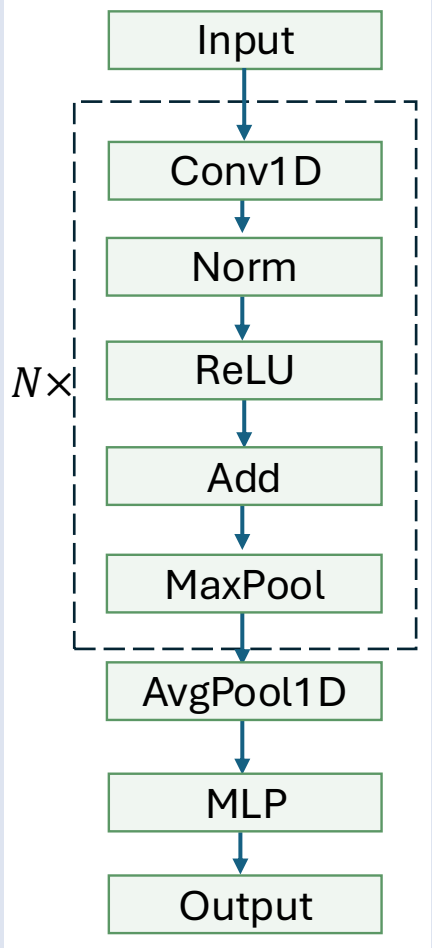
D2 Copresheaf transformer



D3 Artificial neural network



D4 Convolutional neural network



E Consensus prediction

Modality-architecture base models

